Università degli Studi di Parma
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[redacted]	[redacted]	[redacted]

Cybersecurity
Exam 22/4/2023

1) **What is a *known-plaintext* attack?**
[redacted]
☐ An a attack scheme to a cryptographic system where the attacker can choose some messages m_i and, for each m_i, he can capture the corresponding ciphertext $c_i=E_K(m_i)$
☐ An a attack scheme to a cryptographic system where the attacker can choose some ciphertext messages c_i and, for each c_i, she can obtain the corresponding cleartext $m_i=D_K(c_i)$
☐ An a attack scheme to a cryptographic system where the attacker can capture some ciphertext messages $c_i=E_K(m_i)$ and, for each c_i, he can obtain the corresponding cleartext m_i

DSA is:
☐ A symmetric block cipher algorithm
☐ An asymmetric block cipher algorithm
☐ A hash algorithm
☒ A signature algorithm

2) **Given a block cipher E in CBC mode, with block size *r* and key length *n*, let (m_i,c_i) a pair of plaintext and ciphertext, with message length *tr*. Which is the complexity of a brute force attack against the secret key, in terms of number of single block encryption operations, supposing that in each attempt the entire message is processed.**
[redacted]
☐ $n\,2^r$
☐ $t\,r\,n$
☒ $tr\,2^n$
☐ $t\,2^n$

What do you need in order to verify the validity of a digital certificate?
[redacted]
☐ the certificate of the CA that signed the given certificate
☐ the private key of the CA that signed the given certificate
☐ your own certificate

4) **Which is the advantage to use sequence numbers, timestamps, or random values, in an authentication scheme?**
[redacted]
☐ to protect against brute force attack
☐ to protect against replay or reflection attacks
☐ to increase the security, in case of password too short
☐ to distinguish request messages from responses

5) **Which service is NOT provided by IPSec ESP?**
[redacted]
☐ confidentiality
☐ delivery confirmation
☐ data integrity check
☐ protection against replay attacks

6) **What is a Spoofing attack?**
[redacted]
☐ listening on the transmission medium along the path between two entities that are communicating
☐ forcing a system (the victim) to send the data unencrypted rather than encrypted
☐ sending data with a false sender address, impersonating the identity of another system

7) **Let us consider the following cleartext:**

m = 1100 0000 1100 0000

It is sent encrypted with a symmetric block encryption algorithm $E_K(\cdot)$, with block size equal to 4bit, with key K, using OFB concatenation mode with IV=0001. The complete substitution table of $E_K(\cdot)$ with key K is reported her on the right. The resulting cipher text is:

plaintext	ciphertext
0000	1110
0001	0100
0010	1101
0011	000
0100	00
0101	1
0110	
0111	
1000	
1001	
1010	
1011	
1100	
1101	
1110	
1111	

c = 1000 0010 0001 1001 (IV=0001)

Please indicate the value of the modified ciphertext c' and IV' such as the corresponding cleartext (when decrypted with the same key K) will be:

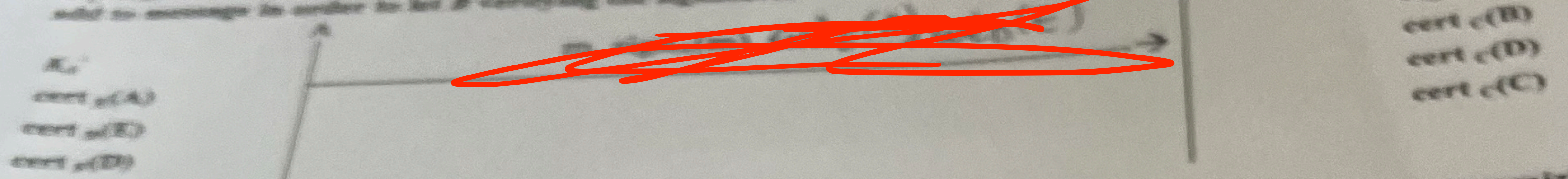
m'= 1100 0000 1001 0000

(IV' = [redacted])

Response: c = [redacted]

[redacted]

13) An entity A wants to send a message m to B signed with her private key K_A^-. Consider that A has her own private key K_A^-, the cert$_E$(A), cert$_E$(E), and the cert$_E$(D), while B has cert$_C$(B), cert$_C$(D), cert$_C$(C). Which information should A add to message in order to let B verifying the signature?



[redacted]

14) The entity A wants to anonymize a message m to be sent to D, by using the cascade of high-latency anonymizing Mix nodes B and C, in such a way that A→B→C→D is the sequence of nodes that will be involved in the delivery. Assume that K^+_i and K^-_i are the public and private keys of node i (with i=A,B,C,D). What is a possible message that A will send to B for such a purpose?

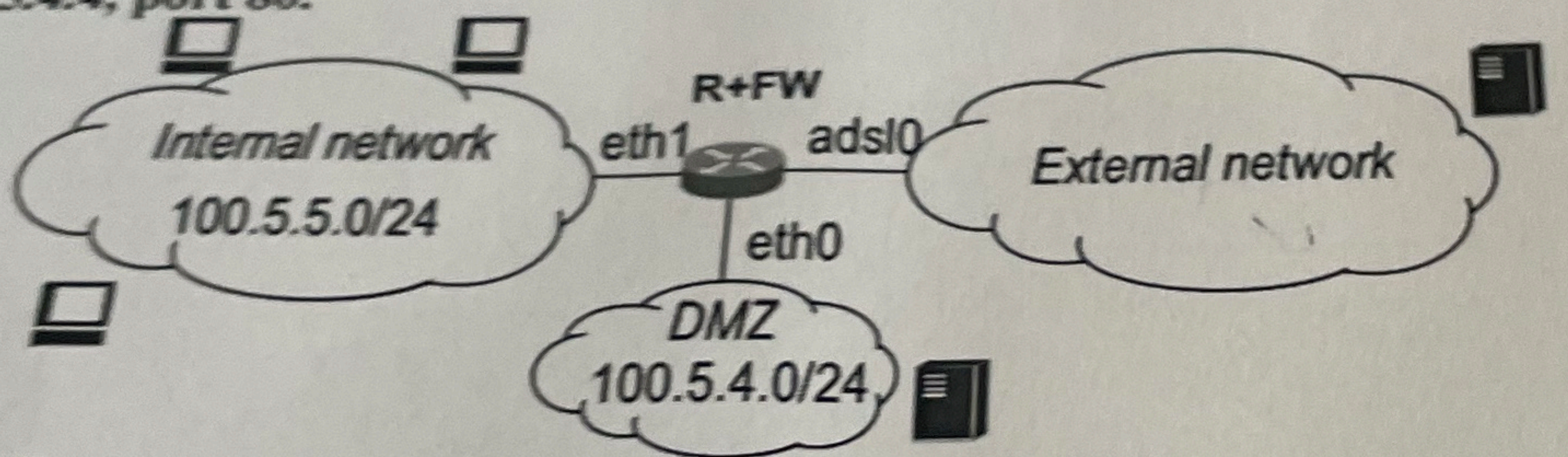
[redacted]

15) Let us consider the following network topology. You are requested to configure the packet filtering router (FW) so that:

a) it is possible to communicate in HTTP from any client of the internal network to any server of the external network, limited to server port 80;

b) it is allowed any communication between any node of the internal network and DMZ;

c) it is possible to establish client/server communications from any node (client) of the external network to the HTTP server 100.5.4.4, port 80.



FORWARD

matching						action
src_addr	dest_addr	proto	src_port	dest_port	conn_state (NEW / ESTABLISHED)	ACCEPT/ DROP
[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

8) What is Euler's totient function?

[redacted]

9) Show a possible challenge-response authentication scheme that can be used by Carol to authenticate David, based on a digital signature algorithm using the private/public keys of Carol and/or David.

Carol

David

[redacted]

[redacted]

10) Create a pair of RSA public/private key pair $K^+=<e,n>$ (public) and $K^-=<d,n>$ (private), starting from the two secret prime numbers $p=3$, $q=19$, and value $d=23$. For obtaining the value e of the public key, you can either use the Euclid's algorithm or try and test knowing that e is lesser than 20. By using the public key K^+ do encrypt the plaintext $m=2$.

[redacted]

11) An entity A wants to send a message m to B, by guaranteeing ONLY the confidentiality of the data (i.e. the message m). For the encryption of m A uses a symmetric encryption algorithm. If A and B do not share in advance any symmetric key, if both A and B have an own RSA private key (respectively K_A^- and K_B^-), and if they share the corresponding public keys K_A^+ and K_B^+, please indicate a possible scheme used by A to send the message and the scheme used by B to receive it:

Sender side	Receiver side
[redacted]	[redacted]

Data actually sent:

12) Show the Diffie-Hellman exchange between Carol and David, using the generator $g=2$, the prime $p=13$, and the values $x_C=4$ and $x_D=5$ as secret values of Carol and David, respectively. Indicate the exchanges values y_C and y_D, and the resulting DH secret computed separately by Carol and David.

[redacted]